

**Cell Type** Bacterial, gram negative

**Molecules Electroporated** DNA: plasmid from 2kb to 30 kb from a ligation mix, sometimes supercoiled.

**Species Used** *E. coli*, DH5α

### Before the Pulse

**Cell Growth Medium** LB

**Growth Phase at Harvest** O.D. (600) = between 0.4 and 0.6

**Pre-pulse Incubation** 30 seconds to 1 minute

**Wash Solution** double distilled water (ice cold) and (0°C, ice cold) glycerol

### The Pulse

**Instruments Used** Gene Pulser® apparatus and Pulse Controller

**Electroporation Temperature** 0 °C

**Electroporation Medium** 10% glycerol/water

**Cuvette Gap** 0.2 cm

**Voltage** 2.5 kV

**Cell Density** Not given

**Volume of Cells** 40 µl

**Field Strength** 12.5 kV/cm

**DNA Concentration** See notes

**DNA Resuspension Buffer** Not given

**Capacitor** 25 µF

**Volume of DNA** from 0.5 µl to 2 µl

**Resistor** 200 Ω (Pulse Controller)

### After the Pulse

**Time Constant** Not given

**Outgrowth Medium** SOC

### Relevant Publications and/or Comments

**Note:** exponential values designated in parentheses.

**DNA concentration:** Depends on ligation-usually straight out of gel slice (low melt) ligation.

**SOC:** 2% Bacto tryptone, 0.5% Bacto yeast extract, 10mM NaCl, 2.5mM KCl, 10 mM MgCl<sub>2</sub>, 10 mM MgSO<sub>4</sub>, 20 mM glucose.

**LB:** 1% Bacto tryptone, 0.5% Bacto yeast extract, 0.5% NaCl.

**Outgrowth Temperature** 37 °C

**Length of Incubation** 20 min. to 1 hour

**Selection Method or Assay Used** Have used carpenicillin, kanamycin, tetracycline, Xgal / IPTG (depending on plasmid)

**Electroporation Efficiency** 10 (9) transformants / µg DNA

**Per Cent Survival** Not given

**Name of Submitter** Not given

**Institution Address**

**Survey Number**

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